HIROFUMI INAGUMA

Speech and Audio Processing Laboratory, Kyoto University, Sakyo-ku, Kyoto 606-8501, Japan
Office:(+81)75-753-5992 ⋄ Moble: (+81)80-6967-1623
inaguma@sap.ist.i.kyoto-u.ac.jp / hiro.mhbc@gmail.com

RESEARCH INTERESTS

Automatic speech recognition
- End-to-end speech recognition
- Multilingual end-to-end speech recognition
- Language modeling
- Transfer learing, domain adaptation
- Low-resource speech recognition

Speech translation
- End-to-end speech translation
- Multilingual end-to-end speech translation

EDUCATION

Ph.D. in Computer Science, Kyoto University, Kyoto, Japan *April 2018 - Present*
Department of Intelligence Science and Technology, Graduate School of Informatics
- Supervisor: Prof. Tatsuya Kawahara

M.Sc. in Computer Science, Kyoto University, Kyoto, Japan *April 2016 - March 2018*
Department of Intelligence Science and Technology, Graduate School of Informatics
- Thesis title: Joint Social Signal Detection and Automatic Speech Recognition based on End-to-End Modeling and Multi-task Learning
- Supervisor: Prof. Tatsuya Kawahara

B.Sc. in Computer Science, Kyoto University, Kyoto, Japan *April 2012 - March 2016*

WORK EXPERIENCES

Microsoft Research, Redmond, WA, USA *July 2019 - October 2019*
Research Internship (3 months)
- Mentor: Dr. Yifan Gong

Johns Hopkins University, Baltimore, MD, USA *July 2018 - September 2018*
Visiting student (2.5 months)
- Worked on end-to-end speech recognition and translation
- Participated in the JSALT workshop (topic: multilingual end-to-end speech recognition)
- Participated in IWSLT2018 end-to-end speech translation evaluation campaign
- Mentor: Prof. Shinji Watanabe

IBM research AI, Tokyo, Japan *September 2017 - November 2017*
Research Internship (2 months)
- Worked on end-to-end ASR systems
- Mentor: Dr. Gakuto Kurata, Dr. Takashi Fukuda

Recruit Co.,Ltd., Tokyo, Japan *February 2017*
Research Intern (2 weeks)
- Worked on data analysis and image classification competition
- Won the 1st place

CyberAgent, Inc., Tokyo, Japan *September 2016 - October 2016*
Research Intern (1 months)
- Worked on the development of music recommendation systems

TECHNICAL STRENGTHS

Programming language	Python, Bash, C/C++, Java, LaTeX, PHP, Javascript, CSS
Software & Tools	Espnet, Kaldi
Deep learning framworks	Pytorch, Tensorflow, Chainer

LANGUAGE SKILL

Japanese (native), English (fluent)

AWARDS

Yamashita SIG Research Award, from Information Processing Society of Japan (IPSJ), March 2019
- Paper title: "An End-to-End Approach to Joint Social Signal Detection and Automatic Speech Recognition"

Yahoo! JAPAN award (best student paper), from SIG-SLP, June 2018

Research Fellowship for Young Scientists (DC1), from Japan Society for the Promotion of Science (JSPS), April 2018 - March 2021

Student award, from the Acoustical Society of Japan (ASJ), March 2018

Student award, from the 79th of National Convention of Information Processing Society of Japan (IPSJ), March 2017

RESEARCH PROJECTS

Joint social signal detection and automatic speech recognition
Incorporated social signal detection (e.g., laughter, filler, back-chennels, and disfluencies) into the end-to-end ASR paradigm, and proposed a unified framework for both tasks.

Acoustic-to-ward sequence-to-sequence speech recognition without OOV
Proposed a sequence-to-sequence ASR framework which directly outputs words, and resolved the out-of-vocabulary (OOV) problem by referring to the character-level hypothesis via multi-task learning with the character-level recognition task.

Multilingual end-to-end speech recognition
Developed the language-independent end-to-end ASR system based on a single sequence-to-sequence model, and successfully adapted to unseen languages.

Multilingual end-to-end speech translation
Integrated an effective multilingual framework into the end-to-end speech translation task, and showed significant improvements of the translation performances compared to the conventional bilingual models.

PUBLICATIONS (REVIEW PAPER)

Hirofumi Inaguma, Jaejin Cho, Murali Karthick Baskar, Tatsuya Kawahara, and Shinji Watanabe, "Transfer Learning of Language-Independent end-to-End ASR with Language Model Fusion", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019. (Acceptance Rate: 1774/3815=**46.5**%)

Jaejin Cho, Shinji Watanabe, Takaaki Hori, Murali Karthick Baskar, **Hirofumi Inaguma**, Jesus Villalba, Najim Dehak, "Language Model Integration Based on Memory Control for Sequence to Sequence Speech Recognition", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019.

Hirofumi Inaguma, Masato Mimura, Shinsuke Sakai, and Tatsuya Kawahara, "Improving OOV Detection and Resolution with External Language Models in Acoustic-to-Word ASR", IEEE Spoken Language Technology Workshop (SLT), 2018. (Acceptance Rate: 150/257=**58.3**%)

Masato Mimura, Sei Ueno, **Hirofumi Inaguma**, Shinsuke Sakai, and Tatsuya Kawahara, "Leveraging Sequence-to-Sequence Speech Synthesis for Enhancing Acoustic-to-Word Speech Recognition", IEEE Spoken Language Technology Workshop (SLT), 2018.

Hirofumi Inaguma, Xuan Zhang, Zhiqi Wang, Adithya Renduchintala, Shinji Watanabe and Kevin Duh, "The JHU/KyotoU Speech Translation System for IWSLT 2018", 15th International Workshop on Spoken Language Translation (IWSLT), 2018.

Hirofumi Inaguma, Masato Mimura, Koji Inoue, Kazuyoshi Yoshii, and Tatsuya Kawahara, "An End-to-End Approach to Joint Social Signal Detection and Automatic Speech Recognition", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018. (Acceptance Rate: 1406/2830=**49.7**%)

Sei Ueno, **Hirofumi Inaguma**, Masato Mimura, and Tatsuya Kawahara, "Acoustic-to-Word Attention-Based Model Complemented with Character-level CTC-Based Model", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018.

Hirofumi Inaguma, Koji Inoue, Masato Mimura, and Tatsuya Kawahara, "Social Signal Detection in Spontaneous Dialogue Using Bidirectional LSTM-CTC", 18th Annual Conference of International Speech Communication Association (Interspeech), 2017. (Acceptance Rate: 799/1582=**52.0**%)

Hirofumi Inaguma, Koji Inoue, Shizuka Nakamura, Katsuya Takanashi, and Tatsuya Kawahara, "Prediction of Ice-breaking between participants using prosodic features in the first meeting dialogue", International Conference Multimodal Interaction workshop on Advancements in Social Signal Processing for Multimodal Interaction (ASSP4MI), 2016.

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